

Lab # 02

Introduction to Mini-worlds, Entities, and Attributes



Database Systems Lab (CL-2005)

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Section: BCS-4E

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Introduction to Mini-worlds, Entities, and Attributes

Objectives:

- Understand the concept of a mini-world and its role in database design.
- Identify entities within a miniworld.
- Recognize and list attributes of entities.
- Extract entities from a given real-world scenario.
- Determine attributes for identified entities based on relevance.
- Develop foundational skills to model real-world information on paper.
- Build the ability to distinguish between entities and attributes.

Introduction to Miniworld

A miniworld is a small part of the real world that we want to represent in a database. It contains objects, events, and relationships that are relevant to the problem we are studying.

Think of it as a “snapshot” of reality that you are interested in.

Example: A school, a library, a bank, or a hospital.

Key Points:

- Miniworlds include only the relevant information for a specific problem.
- Anything outside this focus is ignored.
- Miniworlds help us design a database model before creating actual databases.

Miniworlds Examples:

Miniworlds	What it can include?	What it cannot include?
A School	Students, Teachers, Classes, Subjects.	Cafeteria menu, Playground equipment.
A Library	Books, Members, Borrowing Transactions.	Library building layout, Cleaning schedule.
A Hospital	Patients, Doctors, Appointments, Medicines.	Hospital cafeteria, Parking lot.
A Bank	Accounts, Customers, Transactions, Loans.	Bank building design, Security camera footage.

Reasons for Exclusions:

- **A School:** Cafeteria menu and playground equipment are excluded because the focus is on academic records, not facilities.
- **A Library:** Building layout and cleaning schedules are excluded because only the data related to books and members is needed.
- **A Hospital:** Cafeteria and parking lot are excluded because only patient and treatment information is relevant.

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- **A Bank:** Building design and security footage are excluded because the database focuses on accounts, transactions, and customers.

Introduction to Entities

An entity is an object or event in a miniworld about which information is stored. Entities are usually things or events we want to track in a database. Entities are represented as tables in a database.

Characteristics of an Entity:

- **Represents a real-world object or event:** An entity can be a person, place, thing, or event that exists in the miniworld and about which we want to store information.
Example: Student, Book, or Teacher.
- **Usually a noun:** Entities are generally nouns because they refer to objects, people, places, or events.
Example: “Student” can a entity, but “study” (a verb/action) is not.
- **Has a unique identity:** Each entity can be uniquely identified so we can distinguish it from others.
Example: Two students may have the same name, but each student has a unique ID to identify them in the database.
- Not everything in a miniworld is an entity – only things relevant for data storage.

Activity # 01 – Identifying Entities from Miniworlds

Activity:

Write 2 examples of miniworlds you encounter in daily life and list 3–5 things included in each.

Example Solution:

- Grocery Store → Items, Customers, Cashiers, Bills, Orders.
- Gym → Members, Trainers, Equipment, Membership Plans.

Activity # 02 - Identifying Entities from Real-World Scenarios

Scenario:

In a bank, customers have accounts. Employees help customers open accounts and process transactions. The bank has a building, a cafeteria, security cameras, and parking spaces. Customers deposit and withdraw money. Employees manage loans and maintain records.

Example Solution:

Entity/Not Entity	Justification
Customer (Entity)	Real-world object whose information is stored (name, ID, address, etc.)
Account (Entity)	Object representing a customer’s funds; we store details like balance, account type, account number.
Employee (Entity)	Person managing operations; info stored like employee ID, name, designation.
Transaction (Entity)	Event of deposit or withdrawal; tracked in database.
Building (Not an Entity)	Physical infrastructure; we don’t store building details in this miniworld database.
Cafeteria (Not an Entity)	Operational detail; not relevant to banking data storage.

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Security Cameras (Not an Entity)	Equipment; irrelevant to database tracking of customers and accounts.
Parking Spaces (Not an Entity)	Facility detail; not relevant to customer/account info.

Scenario 2 – Class Task 01

A library keeps detailed records of books, members, and librarians. Members borrow and return books, reserve books that are currently unavailable, and sometimes request new books. Librarians manage book inventory, track overdue books, and maintain membership records. The library also records when books are repaired or replaced. The library building includes a reading hall, a coffee corner, computer stations for catalog searches, and cleaning staff. Books have titles, authors, ISBN numbers, genres, and publication years. Members have names, membership IDs, contact details, and membership types. Librarians have employee IDs, names, departments, and work schedules. Occasionally, the library hosts book club events, guest lectures, and exhibitions, which are organized by librarians.

Student Task: Identify the entities and non-entities in the scenario and explain why each item is included or excluded.

Introduction to Attributes

Attributes are properties or characteristics of an entity. They tell us what we want to know about the entity. An attribute tells us more about an entity. Without attributes, an entity is just a name.

For example, “Student” alone doesn’t tell us much, but Name, Age, Class, Roll Number give meaningful information.

Attributes are columns in a table. Each attribute stores a specific piece of information for every entity instance.

For instance, in a “Student” table:

- Columns → Attributes (Name, Roll Number, Age, etc.)
- Rows → Each individual student.

Key Points:

- Attributes are details or properties of an entity.
- They tell us what we want to know about the entity.
- They only include information that is useful or important in a particular environment.
- Each attribute stores one type of information.
- Some attributes are simple (like Age) and some can be combined (like Full Name = First Name + Last Name) and some attributes can be calculated from others (like Age from Date of Birth).
- Attributes can be mandatory (must have a value) or optional.

Data Types for Attributes

A data type is a property of an attribute that defines what kind of information it can store in a database. It tells the database whether the attribute should hold text, numbers, dates, or other types of values.

A data type defines the type of values an attribute can store and the kind of operations that can be performed on it. Every attribute in a database has a data type that helps organize data properly, ensures accurate calculations, and prevents storing wrong information. Choosing the correct data type is an important step in designing a database.

Some Common Data Types:

Data Type	Example	Description
Text / String	Name = "Ali"	Stores letters, words, or sentences.
Number / Integer	Age = 18	Stores whole numbers only.
Decimal / Float	Marks = 87.5	Stores numbers with decimals.
Date	Date of Birth = 20-SEP-1999	Stores dates in day/month/year format.
Boolean	IsMember = Yes	Stores True/False or Yes/No values.

Activity # 03 - Association of Data Types with Attributes

Scenario:

A small educational institute maintains records of its students, books, and members. Students are enrolled in the institute and are identified by their name, roll number, and date of birth. The institute also stores information about books, including their titles, authors and prices. Members (such as staff who are allowed to use institute facilities) are tracked by the system to determine whether they are currently active or inactive.

Example Solution:

Entity	Attributes	Data Types	Justification	Example
Student	Name	Text/String	Names consist of letters and words.	"Sadiq"
	Roll Number	Number/Integer	Roll numbers are mostly numeric values.	129
	Date of Birth	Date	Stores calendar date information.	"12-AUG-1998"
Book	Title	Text/String	Book titles are textual in nature.	"Harry Potter"
	Price	Decimal/Float	Price may include decimal values.	499.98
	Author	Text/String	Author name consists of letters and words.	"J.K Rowling"
Member	MemberID	Number/Integer	MemberID is a numeric value.	7817
	Name	Text/String	Names consist of letters and words	"Shaheer"
	isActive	Boolean	Indicates whether a member is active or inactive.	Yes

Activity # 04 - Designing a Complete Database from a Miniworld

Scenario

A small community art center wants to keep track of its activities, participants, and instructors. The center records workshops, including workshop names, topics, duration, and fees, as well as art supplies available, including their names, quantities, and costs. Participants are registered with a unique participant ID, name, and membership status, while instructors have employee IDs, names, and specialty areas. Although the center also has a gallery, a café corner, and notice boards, these are not recorded in the system, as they are not needed for managing data. This system helps the center efficiently organize workshops, track supplies, monitor participants' attendance, and manage instructors' assignments.

Step – I: Identification of Miniworld.

The miniworld is the Community Art Center system, the part of the real world we want to manage using a database.

Step – II: Identification of Entities.

Entity	Reason
Workshop	Real-world activity with multiple pieces of information (name, topic, duration, fees).
Art Supply	Physical items used in workshops with important details (name, quantity, cost).
Participants	People enrolled in workshops.
Instructor	People conducting workshops.

Step – II: Identification of Attributes.

Entity	Attributes
Workshop	WorkshopName, Topic, Duration, Fee.
Art Supply	SupplyName, Quantity, Cost.
Participants	ParticipantID, Name, MembershipStatus.
Instructor	EmployeeID, Name, SpecialtyArea.

Step – IV: Assignment of Data Types.

Now that we have identified all the entities and their attributes, the next step is to decide what type of data each attribute should store. Every attribute in a database needs a data type, which tells the system whether it is storing text, numbers, dates, or simple yes/no information. Choosing the correct data type is important because it helps the system store data efficiently, prevent errors, and perform calculations or comparisons correctly. For example, storing a participant's name as a number would not make sense, and storing a workshop fee as text would make it difficult to calculate totals.

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Entity	Attributes	Data Type	Reason
Workshop	Workshop Name	Text / String	Contains words.
	Topic	Text / String	Textual description.
	Duration	Number / Integer	Duration in hours (numeric).
	Fee	Decimal / Float	Price may include decimals.
Art Supply	Supply Name	Text / String	Contains words.
	Quantity	Number / Integer	Quantity is numeric.
	Cost	Decimal / Float	Cost may include decimals.
Participant	Participant ID	Number / Integer	Unique ID to identify participant.
	Name	Text / String	Contains letters and words.
	Membership Status	Boolean	Indicates active membership.
Instructor	Employee ID	Number / Integer	Unique ID to identify instructor.
	Name	Text / String	Contains letters and words.
	Specialty Area	Text / String	Describes area of expertise.

Final Note: Physical things like gallery, café corner, notice boards are not entities, because the system doesn't store information about them.

Always start by understanding the real-world scenario carefully. Focus on identifying only the important entities and their attributes, and choose data types that make sense for the information you want to store. Thinking clearly at this stage will make designing databases and writing queries much easier in the future.

THE END!